

Abstract No.

Transcriptomic signature and cellular plasticity of glioblastoma circulating tumor cells

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Background

Glioblastoma (GBM) remains the most aggressive and lethal primary brain tumor in adults. The presence of circulating tumor cells (CTCs) in the peripheral blood has been well-documented. CTC-based liquid biopsy represents a promising, non-invasive tool for longitudinal disease monitoring.

Methods

Peripheral blood samples were collected from 4 patients diagnosed with GBM. CTCs were enriched using the Parsortix microfluidic platform. To eliminate false positives arising from background leukocyte contamination and microfluidic-induced shear stress, we performed RNA-sequencing (RNA-seq) in parallel on both the enriched CTC pool and matched background blood cells (BC) from the same patient, subjected to the exact same Parsortix protocol. Differential gene expression analysis ($\log_2$ Fold Change > 1) and functional enrichment analysis (MSigDB Hallmark 2020) were performed by comparing CTCs against their matched BC controls (CTCs vs BC) to isolate a pure CTC signature.

Results

The patient-matched background subtraction strategy successfully isolated highly specific transcriptomic alterations in GBM CTCs. MSigDB Hallmark pathway analysis revealed a robust and statistically significant enrichment in Epithelial-Mesenchymal Transition (EMT) ($p = 3.72 \times 10^{-4}$), KRAS Signaling Down ($p = 4.96 \times 10^{-3}$), Myogenesis, and Coagulation pathways in CTCs. Conversely, CTCs demonstrated a significant down-regulation of pathways associated with the inflammatory response and TNF-alpha Signaling via NF-kB ($p = 2.50 \times 10^{-5}$).

At the single-transcript level, CTCs exhibited an up-regulation of stem/proneural markers such as SOX2, classical GBM markers including GFAP and S100B. In contrast, prominent mesenchymal markers were either down-regulated or unchanged (e.g., CHI3L1, SERPINE1), suggesting a distinct state of fluid-phase cellular plasticity.

Conclusions and Future Perspectives

These preliminary data highlight the activation of specific survival and plasticity programs (EMT) during hematogenous dissemination. Ongoing work focuses on directly comparing this clean circulating signature with the primary solid tumor transcriptome of the same patients to map clonal evolution, tumor heterogeneity, and mechanisms of therapy resistance.



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